

Quiz (Carolina Reaper)

Q1. Simplify $\sqrt{48} + \sqrt{18}$

- (A) $4\sqrt{3} + 3\sqrt{2}$
- (B) $4\sqrt{2} + 3\sqrt{3}$
- (C) $2\sqrt{6} + 3\sqrt{2}$
- (D) $4\sqrt{3} + 9$
- (E) $\sqrt{66} + \sqrt{2}$

Q2. Simplify $\frac{\sqrt{50}}{\sqrt{2}}$

- (A) 5
- (B) $\sqrt{25}$
- (C) $\sqrt{10}$
- (D) $\frac{5}{\sqrt{2}}$
- (E) $\frac{\sqrt{25}}{\sqrt{2}}$

Q3. A shipping truck travels 45 miles in 3 hours. If the truck's speed is increased by a factor of $\sqrt{2}$, how many miles will it travel in 2 hours?

- (A) $45\sqrt{2}$
- (B) 90
- (C) $45 \cdot \sqrt{2} \cdot \frac{2}{3}$
- (D) $30\sqrt{2}$
- (E) $60\sqrt{2}$

Q4. Multiply $\sqrt{12} \cdot \sqrt{3}$

- (A) 6
- (B) $\sqrt{36}$
- (C) $\sqrt{9} \cdot \sqrt{4}$
- (D) $2\sqrt{9}$
- (E) $6\sqrt{2}$

Q5. A logistics company has two delivery routes. Route A takes

$2\sqrt{5}$ hours, and Route B takes $\sqrt{20}$ hours. Which route takes longer?

- (A) Route A
- (B) Route B
- (C) Both take the same time
- (D) Need more information
- (E) Route B is $\sqrt{2}$ times longer

Q6. Simplify $\frac{\sqrt{8}}{\sqrt{2}}$

- (A) $\sqrt{4}$
- (B) 2
- (C) $\sqrt{2}$
- (D) $\frac{2}{\sqrt{2}}$
- (E) $\frac{\sqrt{4}}{\sqrt{2}}$

Q7. Subtract $\sqrt{75} - \sqrt{48}$

- (A) $\sqrt{3}$
- (B) $\sqrt{27}$
- (C) $\sqrt{25} - \sqrt{16}$
- (D) $\sqrt{3} \cdot (\sqrt{25} - \sqrt{16})$
- (E) $5\sqrt{3} - 4\sqrt{3}$

Q8. A traffic flow model predicts the number of cars passing a point on the highway is $\sqrt{t} \cdot 2$ in t minutes. How many cars will pass the point in 9 minutes?

- (A) $3 \cdot 2$
- (B) $\sqrt{9} \cdot 2$
- (C) $\sqrt{18} \cdot 2$
- (D) 6
- (E) $6 \cdot \sqrt{2}$

Open Ended Questions (FRQ)

Q9. A delivery truck travels from Warehouse A to Warehouse B. If the distance between the two warehouses is d miles and the truck's speed is $s = \sqrt{t}$ miles per hour, where t is the time in hours, derive an equation for the time it takes the truck to travel from Warehouse A to Warehouse B.

Q10. Simplify $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b})$ and show all steps. Assume a and b are positive numbers.

Chef's Tips

- Double-check student work for the correct application of radical rules.
- Encourage students to simplify radical expressions whenever possible.
- Consider using real-world scenarios to make radical operations more engaging.